Q1. Write short note on registration, tunneling, and Encapsulation

Ans:

Registration:

Registration is the process where a mobile node informs its home network about its current location. When the device moves to a foreign network, it obtains a Care-of Address and sends a registration request through the Foreign Agent to the Home Agent. The Home Agent then updates the mobile node's location so that data can be delivered correctly.

Tunneling:

Tunneling is the method used to transfer data packets from the home network to the mobile node in a foreign network. In this process, the Home Agent creates a virtual path and sends packets to the Care-of Address. This allows communication even when the node is away from its home network.

Encapsulation:

Encapsulation is the process of wrapping the original data packet inside another packet by adding a new header. This is done by the Home Agent, where the new header contains the Care-of Address as the destination. It is mainly used to support tunneling and ensure correct delivery of packets.

Q2.Explain packet delivery in mobile IP

Ans:

Packet delivery in Mobile IP works indirectly through the home network. When a correspondent node sends data, it is first delivered to the Home Agent using the mobile node's home address. The Home Agent then finds the current Care-of Address and sends the packet through tunneling by encapsulating it. In the foreign network, the packet is decapsulated and delivered to the mobile node. The reply from the mobile node is usually sent directly back to the sender, which creates triangular routing.

Packet delivery in mobile IP Working:

Step 1: The correspondent node sends a packet to the home address.

Step 2: The Home Agent receives it and finds the Care-of Address.

Step 3: The Home Agent encapsulates and tunnels the packet.

Step 4: The packet reaches the foreign network and is decapsulated.

Step 5: The mobile node receives the packet.

Step 6: The reply is sent directly to the correspondent node.

Q3.Explain DHCP and TCP Snooping

Ans:

DHCP (Dynamic Host Configuration Protocol):

DHCP is used to automatically assign IP addresses and network configuration to devices in a network. When a device connects, it sends a request to a DHCP server, which responds with an IP address, subnet mask, gateway, and DNS. This eliminates manual configuration and ensures efficient network management.

TCP Snooping (Snoop Protocol):

TCP Snooping is a technique used in wireless networks to improve TCP performance. A snoop agent is placed at a base station or access point, where it monitors TCP packets. If packet loss occurs due to wireless errors, the snoop agent retransmits the lost packets locally instead of letting the sender reduce its speed. This avoids unnecessary congestion control and improves overall data transmission efficiency.

Q4.Write the features of 3G,4G,5G,6G

Ans:

3G (Third Generation):

3G provides better voice and data services compared to 2G. It supports internet browsing, video calling, and multimedia services. It offers moderate data speed (up to a few Mbps) and improved connectivity for mobile users.

4G (Fourth Generation):

4G offers high-speed internet and better performance than 3G. It supports HD video streaming, online gaming, and fast downloads. It uses technologies like LTE and provides speeds up to 100 Mbps or more with low latency.

5G (Fifth Generation):

5G provides very high data speed, ultra-low latency, and massive device connectivity. It supports advanced applications like IoT, smart cities, autonomous vehicles, and AR/VR. Speeds can reach up to several Gbps.

6G (Sixth Generation):

6G is still under development and expected to provide extremely high speeds (up to Tbps), near-zero latency, and intelligent communication using AI. It will support advanced technologies like holographic communication, smart environments, and highly connected systems.

Q5.Define WLAN and List and explain types of WLAN

Ans:

WLAN (Wireless Local Area Network):

WLAN is a network that connects devices wirelessly within a limited area such as a home, office, or campus using radio waves instead of cables.

Types of WLAN:

1. Infrastructure WLAN:

Devices communicate through a central Access Point, which manages the network and connects devices to the internet. It is commonly used in homes and offices.

2. Ad-hoc WLAN:

Devices connect directly with each other without any Access Point. It is simple and useful for temporary communication but has limited range.

3. Mesh WLAN:

Multiple Access Points are interconnected, allowing data to travel through different paths. This improves coverage and reliability in large areas.

4. Hybrid WLAN:

It combines infrastructure and ad-hoc networks, where some devices use an Access Point while others connect directly, providing flexibility.

Q6.Write short note on bluetooth

Ans:

Bluetooth is a simple wireless technology that lets devices connect and share data over short distances without using cables. It is commonly used to connect phones with headphones, speakers, keyboards, and other devices. It works on the 2.4 GHz frequency and usually covers a range of about 10 meters. Bluetooth is easy to use and mainly used for personal devices.

Advantages:

No wires needed

Low power consumption

Easy to connect and use

Low cost technology

Widely supported by devices

Disadvantages:

Limited range

Slower data speed

Can face interference and security issues

Lower bandwidth compared to Wi-Fi

Connection can drop at longer distance